CSE 4345: Big Data Analytics

Week 1, Part 2 — Research Articles & Case Study

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Part 2 Overview

Today's Agenda

1. Research Paper 1 — Laney (2001): The 3Vs Origin
2. Case Study — McKinsey Global Institute (2011)

Seminal Papers Covered

1. Laney, D. (2001). *3D Data Management*. META Group.

Case Study

Manyika et al. (2011). *Big Data: The Next Frontier for Innovation, Competition, and Productivity*. McKinsey Global Institute.

CLO Addressed

CLO1 — Understand big data characteristics
PLO(a)

Research Paper 1 — Laney (2001): The 3Vs Origin

Laney (2001): Publication Context

Full Citation

Laney, D. (2001, February 6). **3D Data Management: Controlling Data Volume, Velocity and Variety.** META Group Application Delivery Strategies Research Note, File 949.

Venue: META Group analyst memo (2 pages)
Cited by: >20,000 times (Google Scholar, 2024)
Impact: Became the canonical industry and academic definition of Big Data

The Author

Doug Laney was a research analyst at META Group (later acquired by Gartner in 2005). He wrote this as a short practitioner note about **e-commerce data management challenges**—not a formal academic paper.

Laney (2001): Publication Context

Laney's Own Reflection (2012 Interview)

"It was sort of a throwaway piece. I had been thinking about the data challenges at e-commerce companies and wrote down three dimensions. I never expected it to become the definition of an entire era of computing."

—Doug Laney, Gartner Blog, 2012

Why Study a 2-page Memo?

It illustrates that **foundational frameworks** need not come from prestigious venues. The quality of the *insight* matters more than the journal. This is a recurring theme in Big Data research.

Laney (2001): The Original 3Vs—Close Reading

Opening Argument (Paraphrased)

"E-commerce has exploded the volume, velocity and variety of information that companies must manage. Retailers and their trading partners face the challenge of managing large, fast-moving and diverse data from the Web, RFID, point-of-sale sensors and order entry systems."

Laney's original definitions:

- **Volume:** Transactions at an online retailer generate **exponentially more** data than equivalent brick-and-mortar operations. Click-stream data, session logs, product views—each visit generates thousands of records.
- **Velocity:** The **time-sensitivity** of data. Stock prices, fraud signals, and real-time offers must be processed within seconds or the business opportunity is lost.
- **Variety:** Data arriving in **many formats**—structured purchase records, semi-structured web logs, unstructured customer reviews.

Laney (2001): The Original 3Vs—Close Reading

The Insight That Made This Seminal

Laney was the first to observe that these three dimensions are **interdependent and compound**:

- High velocity + high variety = integration is nearly impossible in real time
- High volume + high variety = storage schemas must be flexible
- High volume + high velocity = single-node processing cannot keep up

This **3D problem space** is why no single traditional tool solves Big Data—you need an *ecosystem*.

Laney (2001): The E-Commerce Motivation

The year is 2001. The dot-com boom has just ended. What data problems were practitioners actually facing?

Laney's Observed Challenges

Dimension	**Observed Problem (2001)**
Volume	Amazon.com: millions of product pages, billions of click-stream events per month. No RDBMS could store it all.
Velocity	eBay: real-time auction bids must update in seconds. Batch-nightly processing was useless.
Variety	RFID supply-chain data + web logs + customer email = three incompatible formats, no unified query.

Laney (2001): The E-Commerce Motivation

2024 Parallel: Same Problem, 1,000$\times$ Scale

- **Volume (2024):** Amazon: $>$1.6 million orders per day; petabytes of product images, reviews, fulfillment data
- **Velocity:** Daraz.com.bd: flash-sale order spikes of 10,000 orders/minute require real-time inventory sync
- **Variety:** Shopper GPS + purchase history + product image + chatbot transcript + payment gateway log = five different formats in one customer journey

The Enduring Relevance

Laney's 3Vs from 2001 describe a problem that has only **grown more severe**. The framework is still the standard taxonomy in every major cloud provider's documentation (AWS, GCP, Azure) in 2024.

Laney (2001): Legacy, Extensions, and Critique

What the paper triggered:

- Gartner adopted 3Vs as official Big Data definition (2012)
- IBM extended to **4Vs** (adding Veracity, 2013)
- IDC added **Value** as the 5th V (2014)
- Over 40 papers have proposed additional Vs (Validity, Volatility, Venue, Vocabulary. . .)

The 5Vs Consensus (Gartner / IDC, 2012–2014)

V	Added by
Volume, Velocity, Variety	Laney (2001)
Veracity (quality)	IBM (2013)
Value (utility)	IDC (2014)

Laney (2001): Legacy, Extensions, and Critique

Academic Critique of the 3Vs

The 3Vs are a **useful taxonomy**, not a formal theory:

- They describe *symptoms*, not solutions
- They do not specify *thresholds*: how much volume is "big"?
- They ignore *social and ethical* dimensions of data (privacy, consent, bias)
- Kitchin (2014) argues they reinforce a **techno-deterministic** view that more data always produces better insights

Reference: Kitchin, R. (2014). *The Data Revolution*. Sage. Ch. 4.

Takeaway for Practitioners

Use the 3Vs to **diagnose** problems. Do not use them to **justify** adding technology without understanding the specific business value (V5).

Case Study — McKinsey Global Institute (2011)

About the Report

Full Citation

Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Byers, A.H. (2011, May). **Big Data: The Next Frontier for Innovation, Competition, and Productivity.** McKinsey Global Institute.

Length: 156 pages
Downloaded: >2 million times
Cited by: >15,000 academic papers
Lead author: James Manyika, Director, McKinsey Global Institute

Scope

Surveyed Big Data potential across **five domains**: US healthcare, European public administration, US retail, global manufacturing, and personal location data. Included interviews with 250 executives across 14 industries.

About the Report

Why It Mattered in 2011

This was the first large-scale, cross-industry **quantified economic analysis** of Big Data's potential value. Before this report, Big Data was primarily a *technical* term used by engineers. McKinsey made it a *boardroom* term.

Historical Context (2011)

- Hadoop 1.0 had just been released (2011)
- Spark did not yet exist (2012)
- "Data Scientist" was named best job in America by HBR (2012)
- AWS was 5 years old; cloud computing was nascent
- This report **triggered the Big Data hiring wave** of 2012–2016

Key Quantitative Findings

Domain	Estimated Value
US Healthcare	**\$300 billion+/year**
European Public Sector	**€100 billion+/year**
US Retail (operating margin)	**60%+ increase**
Global Manufacturing	**\$600 billion+**
Personal location data (EU)	**\$100 billion+**
US talent shortfall (deep analytics)	**140,000–190,000 people by 2018**
US manager shortfall (data-savvy)	**1.5 million by 2018**

Methodology Note

Values estimated via expert interviews, revenue-impact modelling, and extrapolation from early big data deployments. **Not derived from controlled experiments**—a key limitation noted by critics.

Key Quantitative Findings

The Headline Argument

"Big data will become a key basis of competition, underpinning new waves of productivity growth, innovation, and consumer surplus. Leaders in every sector will have to grapple with the implications of big data, not just a few data-oriented managers."

—Manyika et al., 2011, p. 1

The Talent Gap Was Under-estimated

The report predicted 140,000–190,000 deep analytics jobs needed by 2018. LinkedIn's 2018 Workforce Report found **250,000+ unfilled data science roles** in the US alone—the actual gap was *larger* than McKinsey forecast.

Healthcare: The Flagship Case

US Healthcare: $300B+ Annual Value Potential

McKinsey identified five mechanism categories:

1. **Clinical decision support**: Mining EHR data to identify optimal treatments and reduce diagnostic errors. Potential: $165 billion.
2. **R&D optimisation**: Mining clinical trial data to reduce drug development cost and time. Potential: $70 billion.
3. **Public health**: Epidemic detection via syndromic surveillance of pharmacy sales and emergency room data. Potential: $30 billion.
4. **Fraud and waste detection**: Identifying anomalous billing patterns in Medicare/Medicaid claims. Potential: $17 billion.
5. **Consumer health**: Mining wearable and app data for preventive care.

Healthcare: The Flagship Case

2024 Reality Check: What Was Achieved?

Largely realised:

- Google DeepMind's AlphaFold (2020) — solved protein structure prediction using big data + deep learning
- CMS fraud detection using ML saves $4B/year (2022 GAO report)
- COVID-19 genomic surveillance systems (GISAID) directly applied big data to track variants in real time

Underachieved:

- EHR interoperability remains poor (data silos still exist)
- **HIPAA compliance** slows data sharing; predicted gains require data McKinsey assumed would flow freely

Critical Analysis: What Did McKinsey Get Right and Wrong?

What McKinsey Got Right ✓

- **Talent gap**: understated, not overstated
- **Competitive differentiation**: tech companies that used big data (Netflix, Amazon, Google) dominated their industries
- **Healthcare value**: clinical AI has validated many of the predicted gains
- **Retail personalisation**: Amazon's recommendation engine drives ≈35% of revenue—enabled by big data
- **Urgency**: The hiring and investment wave that followed this report built the infrastructure now used for AI

Critical Analysis: What Did McKinsey Get Right and Wrong?

What McKinsey Missed ×

- **Privacy & regulation**: No mention of GDPR (2018), CCPA, or the political cost of mass data collection (Cambridge Analytica, 2018)
- **Algorithmic bias**: Big data models can *amplify* discrimination (race, gender, credit scoring)
- **Google Flu Trends cautionary tale (2008–2015)**: Hyped as proof of big data power; failed catastrophically due to **overfitting and Goodhart's Law**
- **Cloud dominance**: Assumed on-premise Hadoop clusters; actual market moved to AWS/GCP/Azure
- **Deep learning**: Report assumed statistical models; neural networks redefined "analytics"

Cautionary Interlude: Google Flu Trends (2008–2015)

The Claim (Ginsberg et al., 2009, *Nature*)

"We can accurately estimate influenza activity in each region of the United States, up to two weeks ahead of traditional CDC reports, simply by tracking the volume of flu-related Google search queries."

Initial results: Predicted flu activity with $r^2 = 0.97$ correlation to CDC data. Widely cited as proof that big data could replace traditional surveillance.

What Went Wrong

- **Overfitting**: 45 query terms selected post-hoc from 50 million possibilities
- **Goodhart's Law**: When a measure becomes a target, it ceases to be a good measure. Media coverage of flu *itself* drove search volume.
- **2012–2013**: Model over-predicted flu activity by **2**$\times$
- **2015**: Project quietly discontinued by Google

Cautionary Interlude: Google Flu Trends (2008–2015)

Lessons for Big Data Practitioners

1. **Correlation $\neq$ causation**: A high r^2 on training data does not guarantee real-world validity
2. **Feedback loops**: Human behaviour changes when people know they are being measured
3. **Ground truth matters**: Big data **supplements**, it does not replace, rigorous data collection
4. **Reproducibility**: Selecting 45 features from 50M candidates without cross-validation is a **data dredging** error

Reference: Lazer et al. (2014). *The Parable of Google Flu: Traps in Big Data Analysis.* Science, 343(6176), 1203–1205.

Full References

Seminal Papers

- Laney, D. (2001, Feb 6). **3D Data Management: Controlling Data Volume, Velocity and Variety.** META Group Research Note, File 949.
- Brewer, E. A. (2000). **Towards Robust Distributed Systems.** ACM PODC Keynote, Portland, OR.
- Gilbert, S. & Lynch, N. (2002). **Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services.** *ACM SIGACT News*, 33(2), 51–59.
- Kleppmann, M. (2015). **A Critique of the CAP Theorem.** *arXiv*:1509.05393. *[Recommended reading]*
- Abadi, D. (2012). **Consistency Tradeoffs in Modern Distributed Database System Design: CAP is Only Part of the Story.** *IEEE Computer*, 45(2), 37–42. *[PACELC]*

Case Study & Critique

- Manyika, J. et al. (2011). **Big Data: The Next Frontier.** McKinsey Global Institute.
- Ginsberg, J. et al. (2009). **Detecting influenza epidemics using search engine query data.** *Nature*, 457, 1012–1014.
- Lazer, D. et al. (2014). **The Parable of Google Flu: Traps in Big Data Analysis.** *Science*, 343(6176), 1203–1205.

End of Week 1

CSE 4345: Big Data Analytics

"The world is one big data problem."

—Andrew McAfee, MIT Sloan School

Next Week: Week 2 (Part 1 & Part 2)

Challenges of Big Data: Extraction, Integration & Heterogeneous Information

Seminal Paper: Halevy, Rajaraman & Ordille (2006) · Case Study: Healthcare Interoperability (HL7/FHIR)

Reading before Week 2: [SA] Ch. 2 · **[TE]** Ch. 3